ECON 0100 | Exercise E2 | Competitive Firms

Exercises are a certificate of the work done together in class and offers extra practice outside class.

Q1 | Choosing Quantity

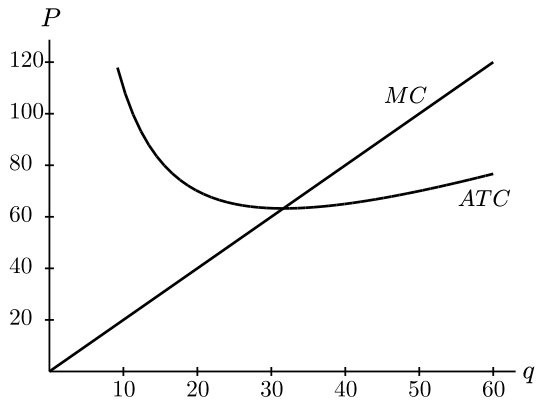
The animation's competitive firm has cost functions: $TC = 1000 + q^2$ and $MC = 2q$.

a) If the market price is $P = 80$, find:

- Total Revenue from selling 20 units: TR_{20}
- Total Revenue selling 21 units: TR_{21}
- Marginal Revenue

b) Calculate and graph the firm's best quantity at $P = 80$.

c) Calculate and graph the firm's maximal profit.
(*Hint: ATC at the profit-maximizing quantity is 65.*)



Q2 | Operating or Leaving

At a price of 40, the firm's best positive output is 20 units. Fixed cost of 1,000 is unavoidable this period but avoidable in future periods by leaving the industry.

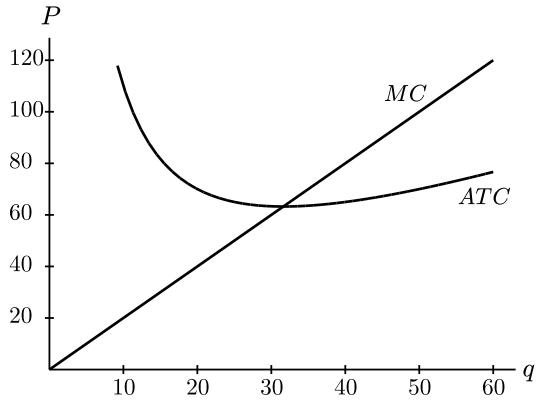
a) Calculate and graph profit from operating this period.
(*Hint: ATC at the profit-maximizing quantity is 70.*)

b) Calculate the profit from shutting down this period.

c) Should the firm operate or shut down this period?

d) In the long run at this price, should the firm exit?

d) In the long run, how does supply and price change?



Q3 | A Higher Marginal Cost

After being in long run equilibrium, suppose every firm's marginal costs go up. Every unit now costs the same additional amount to produce. Use two graphs to examine the impact on the market.

a) What happens to price, each firm's output, and profit after the new cost?

b) What happens to the long run price, each firm's output, market output, and the number of firms?

The Market



One Firm

